A Counterexample for the ℓ_∞^d-Ball Version of Problem 1

Run `fa_banach_001`, lane 0

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Status. Counterexample, likely valid. For every nonempty bounded open $\Omega \subset \mathbb{R}^d$, $d \geq 2$, the James-function space defined using ℓ_∞^d balls contained in Ω contains an isomorphic copy of ℓ_1. Thus the ℓ_∞^d-ball branch of Problem 1(i) in arXiv:1210.2379 has a negative answer. The Euclidean-ball ℓ_1-noncontainment branch is not settled here.

1 Source Problem

The source paper is S. A. Argyros, A. Manoussakis, and M. Petrakis, *Function spaces not containing* ℓ_1, arXiv:1210.2379 [1]. Problem 1 asks: for a family $Q(\Omega)$ of convex bodies contained in a bounded open $\Omega \subset \mathbb{R}^{d_0}$, $d_0 > 1$, when does the space with norm

$$\|f\|_{\mathrm{JF}(Q(\Omega))} = \sup\left\{ \left(\sum_{i=1}^{n} \left(\int_{V_i} f \right)^2 \right)^{1/2} : \{V_i\}_{i=1}^n \subset Q(\Omega),\ V_i \cap V_j = \emptyset \right\}$$

not contain ℓ_1, and when is its dual nonseparable? The authors note that their parallelepiped proof does not yield similar results for other families and state that the problem is open if $Q(\Omega)$ is either the Euclidean balls or the $\ell_\infty^{d_0}$ balls contained in Ω.

Problem 1. Let Ω be an open bounded subset of $\mathbb{R}^{d_0}$, $d_0 > 1$. For $Q(\Omega)$ a family of convex bodies contained in Ω we consider the space $JF(Q(\Omega))$ endowed with the following norm

$$\|f\|_{JF(Q(\Omega))} = \sup\left\{ \left(\sum_{i=1}^{n} (\int_{V_i} f)^2 \right)^{1/2} : \{V_i\}_{i=1}^n \subset (Q(\Omega)), V_i \cap V_j = \emptyset \right\}.$$

For what families $Q(\Omega)$ the following hold:

(i) The space $JF(Q(\Omega))$ does not contains ℓ_1.

(ii) The space $JF^*(Q(\Omega))$ is non-separable.

Remark. Our proof for the family of the paralellepipeds $\mathcal{P}(\Omega)$ does not yield similar results for other families. In particular the above problem is open if $Q(\Omega)$ is either the Euclidean balls or the $\ell_\infty^{d_0}$ balls contained in Ω.

Figure 1: Problem 1 and the ball-family remark in arXiv:1210.2379, PDF page 45.

2 Proof Intuition

The obstruction is that ℓ_∞^d balls are axis-parallel cubes. In dimension at least two, a high-frequency one-dimensional derivative can be spread across the remaining $d-1$ coordinates. Integrating over a cube of side length s produces a factor s^{d-1} from the passive coordinates and an oscillatory primitive difference in the first coordinate. This factor is exactly what keeps the ℓ_2 sum over disjoint cubes uniformly bounded.

At the same time, nested detector cubes inside the support cube read off

$$t^{d-1}\sin(2\pi 2^n t).$$

For every subsequence one can choose a single t whose binary expansion makes these sine values oscillate along a further subsequence. Hence the bounded sequence has no weak-Cauchy subsequence. Rosenthal's ℓ_1 theorem then forces an ℓ_1 subsequence.

3 Counterexample

Let $\Omega \subset \mathbb{R}^d$ be nonempty, bounded, and open, with $d \geq 2$. Let $Q_\infty(\Omega)$ denote the family of open axis-parallel ℓ_∞^d balls, equivalently open cubes with equal side lengths, contained in Ω.

Theorem 1. *The space* $\mathrm{JF}(Q_\infty(\Omega))$ *contains an isomorphic copy of* ℓ_1*. Consequently, the* ℓ_∞^d*-ball case of Problem 1(i) in [1] has a negative answer.*

Proof. Choose an open axis-parallel cube

$$C = (a_1, a_1 + L) \times \cdots \times (a_d, a_d + L)$$

whose closure is contained in Ω. Set

$$\lambda_n = 2\pi 2^n$$

and define $f_n \in L^1(\Omega)$ by

$$f_n(x_1, \dots, x_d) = \begin{cases} \dfrac{\lambda_n}{L}\cos\left(\lambda_n \dfrac{x_1 - a_1}{L}\right), & (x_1, \dots, x_d) \in C, \\ 0, & \text{otherwise.} \end{cases}$$

We first show that (f_n) is bounded in $\mathrm{JF}(Q_\infty(\Omega))$.

Let V be an ℓ_∞^d ball contained in Ω, and let $s(V)$ be its side length. Since $V \cap C$ is a rectangle, writing $s = s(V)$ gives

$$\left|\int_V f_n\right| \leq s^{d-1}\min\left\{2, \frac{\lambda_n}{L}s\right\}.$$

Indeed, the product of the intersection lengths in the last $d-1$ coordinates is at most s^{d-1}, and the first-coordinate primitive

$$\sin\left(\lambda_n \frac{x_1 - a_1}{L}\right)$$

has oscillation at most 2 and derivative bounded by λ_n/L.

Let $V_1, \dots, V_m$ be pairwise disjoint members of $Q_\infty(\Omega)$, and put $s_j = s(V_j)$. Since the cubes are disjoint and contained in the bounded set Ω,

$$\sum_{j=1}^m s_j^d \leq |\Omega|.$$

Let $D < \infty$ be an upper bound for side lengths of cubes contained in Ω. Split the indices according to $s_j \le L/\lambda_n$ and $s_j > L/\lambda_n$.

If $s_j \le L/\lambda_n$, then, since $d \ge 2$ and $\lambda_n \ge 1$,

$$\left| \int_{V_j} f_n \right|^2 \le \left(\frac{\lambda_n}{L} \right)^2 s_j^{2d} \le L^{d-2} \lambda_n^{2-d} s_j^d \le M_1 s_j^d$$

for a constant M_1 depending only on L and d. If $s_j > L/\lambda_n$, then

$$\left| \int_{V_j} f_n \right|^2 \le 4 s_j^{2d-2} = 4 s_j^d s_j^{d-2} \le 4 D^{d-2} s_j^d.$$

Thus

$$\sum_{j=1}^{m} \left| \int_{V_j} f_n \right|^2 \le \max\{M_1, 4D^{d-2}\} \, |\Omega|,$$

uniformly in n and in the finite disjoint cube family. Hence (f_n) is bounded in $\mathrm{JF}(Q_\infty(\Omega))$.

We next show that (f_n) has no weak-Cauchy subsequence. Let (f_{n_k}) be an arbitrary subsequence. Choose a further subsequence, still denoted (f_{m_j}), such that $m_{j+1} - m_j \to \infty$ and $m_{j+1} - m_j \ge 4$.

We claim that there is a $t \in (0, 1)$ such that

$$\sin(2\pi 2^{m_{2j-1}} t) \to 1, \qquad \sin(2\pi 2^{m_{2j}} t) \to -1.$$

Choose the binary expansion of $t = 0.\varepsilon_1 \varepsilon_2 \ldots$ as follows. For odd j, set $(\varepsilon_{m_j+1}, \varepsilon_{m_j+2}) = (0, 1)$; for even j, set $(\varepsilon_{m_j+1}, \varepsilon_{m_j+2}) = (1, 1)$. Between these prescribed blocks put zeros, except where a later prescribed block begins. Since the gaps $m_{j+1} - m_j$ tend to infinity, the fractional part of $2^{m_j} t$ tends to $1/4$ along odd j and to $3/4$ along even j, proving the claim.

For this fixed t, the cube

$$C_t = (a_1, a_1 + tL) \times \cdots \times (a_d, a_d + tL)$$

belongs to $Q_\infty(\Omega)$. Its integration functional is continuous on $\mathrm{JF}(Q_\infty(\Omega))$, and

$$\int_{C_t} f_n = (tL)^{d-1} \sin(2\pi 2^n t).$$

Along the subsubsequence (m_j) these scalar values have two distinct subsequential limits, $(tL)^{d-1}$ and $-(tL)^{d-1}$. Therefore the original subsequence (f_{n_k}) is not weakly Cauchy. Since the subsequence was arbitrary, (f_n) has no weak-Cauchy subsequence.

By Rosenthal's ℓ_1 theorem [2], every bounded sequence in a Banach space has either a weak-Cauchy subsequence or a subsequence equivalent to the unit vector basis of ℓ_1. Applying this to the bounded sequence (f_n), we obtain a subsequence equivalent to the ℓ_1 basis. Hence ℓ_1 embeds into $\mathrm{JF}(Q_\infty(\Omega))$. □

4 Verification and Limitations

The proof is deterministic. The boundedness estimate uses only the displayed definition of the norm and the elementary side-length split at $s = L/\lambda_n$. The no-weak-Cauchy argument uses only cube integration functionals and an explicit binary-expansion selection. Rosenthal's theorem is the standard final step converting a bounded sequence with no weak-Cauchy subsequence into an ℓ_1 basic subsequence.

This packet settles only the ℓ_∞^d-ball branch of the source remark, and it settles that branch negatively for Problem 1(i). It does not address the Euclidean-ball ℓ_1-noncontainment branch. The earlier lane-0 partial packet is preserved inside this packet under the `history/` subfolder, in the directory named `1210.2379_Q_family_nonseparable_dual`; it proves nonseparability of the dual for both Euclidean and ℓ_∞^d ball families. Thus the current combined conclusion for ℓ_∞^d balls is: the dual is nonseparable, but the space contains ℓ_1.

Before promotion, the run indexes were searched for arXiv:1210.2379, the packet slug, and the phrases "Euclidean balls," "ell_infty balls," and "JF(Q)." A bounded web search for arXiv:1210.2379 with "Euclidean balls," "ell_infty," and "James function space" found only the source paper and no obvious later exact resolution of the stated ball-family problem.

5 Human Review Recommendation

The main review point is the interpretation of ℓ_∞^d balls as axis-parallel cubes. Under that standard convention, the construction is direct. Reviewers should also check the binary-selection paragraph and the uniform norm estimate, especially the use of $d \geq 2$ in the small-cube side of the split.

References

[1] S. A. Argyros, A. Manoussakis, and M. Petrakis, *Function spaces not containing* ℓ_1, arXiv:1210.2379, 2012.

[2] H. P. Rosenthal, *A characterization of Banach spaces containing* ℓ_1, Proceedings of the National Academy of Sciences of the United States of America 71 (1974), 2411–2413.